

Note on Bandit Algorithms

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1 Bandits, Probability and Concentration

1.1 Introduction

Definition.

- Actions : $A_1, A_2, \dots \in \mathcal{A}$ and we call $\mathcal{A}$ the action set.
- Rewards : $X_1, X_2, \dots \in \mathcal{R}$.
- History : $H_{t-1} = (A_1, X_1, \dots, A_{t-1}, X_{t-1})$.
- Policy : $\pi \in \Pi : H_{t-1} \mapsto A_t$ and we call Π the competitor class.
- Environment: $\mathcal{E} : A_t \mapsto X_t$.
- Regret of a learner relative to a policy π until $n \in \mathbb{N}$ is

$$R_n = \mathbb{E} \left[\sum_{t=1}^n \mathcal{E}(\pi(H_{t-1})) - \sum_{t=1}^n X_t \right]. \quad (1)$$

where H_0 represents the initial condition.

Example. An environment is called a **stochastic Bernoulli bandit** if

- $\mathcal{A} = \{1, 2, \dots, k\}$,
- $X_t \in \{0, 1\}$,
- $\forall a \in \mathcal{A}, \exists \mu_a \in [0, 1]^k$ such that $\mathbb{P}[X_t = 1 | A_t = a] = \mu_a$.

Here the optimal policy is to play fixed action $a^* = \operatorname{argmax}_{a \in \mathcal{A}} \mu_a$. That is, the competitor class Π is the set of all constant policies $\Pi = \{\pi_1, \dots, \pi_k\}$ where π_i chooses action i in every round. The regret relative to the optimal policy π^* until n rounds is

$$\begin{aligned} R_n &= \mathbb{E} \left[\sum_{t=1}^n \mathcal{E}(\pi^*(H_{t-1})) - \sum_{t=1}^n X_t \right] \\ &= n \max_{a \in \mathcal{A}} \mu_a - \mathbb{E} \left[\sum_{t=1}^n X_t \right]. \end{aligned}$$

Definition. Stochastic stationary bandits are the environments restricted to generate the rewards in response to each action from a distribution independent of the previous history.

Example (Linear Stochastic Bandits).

- $\mathcal{A} \subset \mathbb{R}^d$,
- $X_t = \langle a, \theta \rangle + \eta_t$ for $\theta \in \mathbb{R}^d, a \in \mathcal{A}$ and η_t is a standard Gaussian.

1.2 Foundations of Probability

Definition. Let an event set Ω be given.

- A set $\mathcal{F} \subset 2^\Omega$ is called a **σ -algebra** on Ω if
 - (1) $\Omega \in \mathcal{F}$,
 - (2) $A \in \mathcal{F} \implies A^c \in \mathcal{F}$,
 - (3) $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^\infty A_i \in \mathcal{F}$.
- $\mathcal{G} \subseteq \mathcal{F}$ is a **sub- σ -algebra** if $\mathcal{G}$ is a σ -algebra on Ω .
- $(\Omega, \mathcal{F})$ is called a **measurable space** if $\mathcal{F}$ is a σ -algebra on Ω .
- $\mathcal{B}(\mathbb{R})$ is the **Borel σ -algebra** on $\mathbb{R}$, which is the smallest σ -algebra containing all open intervals (a, b) for $a, b \in \mathbb{R}$.

Definition. Let a measurable space $(\Omega, \mathcal{F})$ be given.

- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is **σ -additive** if
 - (1) $\mu(\emptyset) = 0$,
 - (2) $A_1, A_2, \dots \in \mathcal{F}$ are disjoint $\implies \mu(\bigcup_{i=1}^\infty A_i) = \sum_{i=1}^\infty \mu(A_i)$.
- A function $\mu : \mathcal{F} \mapsto [0, \infty]$ is a **measure** if μ is σ -additive.
- $(\Omega, \mathcal{F}, \mu)$ is called a **measure space** if μ is a measure on $(\Omega, \mathcal{F})$.
- A measure μ is a **probability measure** if $\mu(\Omega) = 1$.
- $(\Omega, \mathcal{F}, \mu)$ is called a **probability space** if μ is a probability measure on $(\Omega, \mathcal{F})$.
- A function $X : \Omega \mapsto \mathbb{R}$ is called a **measurable function** if $\forall B \in \mathcal{B}(\mathbb{R}), X^{-1}(B) \in \mathcal{F}$.

Definition. Let a measure space $(\Omega, \mathcal{F}, \mu)$ be given.

- A function $f : \Omega \mapsto [0, \infty]$ is called a **simple function** if $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ for some $a_1, a_2, \dots, a_n \in [0, \infty]$ and $A_1, A_2, \dots, A_n \in \mathcal{F}$.
- The **integral of a simple function** $f = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ is defined as $\int_\Omega f d\mu := \sum_{i=1}^n a_i \mu(A_i)$.
- The **integral of a non-negative measurable function** $f : \Omega \mapsto [0, \infty]$ is defined as $\int_\Omega f d\mu := \sup_g \int_\Omega g d\mu$ where the supremum is taken over all simple functions g such that $0 \leq g(\omega) \leq f(\omega)$ for all $\omega \in \Omega$.
- The **integral of a measurable function** $f : \Omega \mapsto \mathbb{R}$ is defined as $\int_\Omega f d\mu := \int_\Omega f^+ d\mu - \int_\Omega f^- d\mu$ where $f^+ = \max(f, 0)$ and $f^- = \max(-f, 0)$.
- A function $f : \Omega \mapsto \mathbb{R}$ is called **integrable** if $\int_\Omega |f| d\mu < \infty$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ be given.

- A measurable function $X : \Omega \mapsto \mathbb{R}$ is called a **random variable**.
- The **conditional probability** $\mathbb{P}[A|B] := \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}$ for $A, B \in \mathcal{F}$ and $\mathbb{P}[B] > 0$.
- Two events $A, B \in \mathcal{F}$ are called **independent** if $\mathbb{P}[A \cap B] = \mathbb{P}[A] \cdot \mathbb{P}[B]$.
- Two random variables X, Y are called **independent** if $\forall A, B \in \mathcal{B}(\mathbb{R}), \mathbb{P}[X \in A, Y \in B] = \mathbb{P}[X \in A] \cdot \mathbb{P}[Y \in B]$.

Definition. Let a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a random variable X be given.

- The **expectation** of a random variable X is defined as $\mathbb{E}[X] := \int_{\Omega} X d\mathbb{P}$.
- For a sub- σ -algebra $\mathcal{H} \subseteq \mathcal{F}$, the **conditional expectation** of X given $\mathcal{H}$ is denoted by $\mathbb{E}[X|\mathcal{H}]$ and is defined as a random variable g on Ω such that for all $H \in \mathcal{H}$,

$$\int_H g d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem (2.11). *If $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space and X is an integrable random variable and $\mathcal{H} \subseteq \mathcal{F}$ is a sub- σ -algebra, there exists a unique (almost surely) random variable $\mathbb{E}[X|\mathcal{H}]$ such that for all $H \in \mathcal{H}$,*

$$\int_H \mathbb{E}[X|\mathcal{H}] d\mathbb{P} = \int_H X d\mathbb{P}.$$

Theorem (2.12). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, $\mathcal{G}, \mathcal{G}_1, \mathcal{G}_2 \subseteq \mathcal{F}$ be sub- σ -algebras of $\mathcal{F}$, X, Y be integrable random variables on $(\Omega, \mathcal{F}, \mathbb{P})$. Then the following properties hold:*

- (1) *If $X \geq 0$ almost surely, then $\mathbb{E}[X|\mathcal{G}] \geq 0$ almost surely.*
- (2) *$\mathbb{E}[1|\mathcal{G}] = 1$ almost surely.*
- (3) *$\mathbb{E}[X + Y|\mathcal{G}] = \mathbb{E}[X|\mathcal{G}] + \mathbb{E}[Y|\mathcal{G}]$ almost surely.*
- (4) *If Y is $\mathcal{G}$ -measurable and $\mathbb{E}[XY] < \infty$, then $\mathbb{E}[XY|\mathcal{G}] = Y\mathbb{E}[X|\mathcal{G}]$ almost surely.*
- (5) *If $\mathcal{G}_1 \subseteq \mathcal{G}_2$, then $\mathbb{E}[X|\mathcal{G}_1] = \mathbb{E}[\mathbb{E}[X|\mathcal{G}_2]|\mathcal{G}_1]$ almost surely.*

1.3 Stochastic Processes and Markov Chains

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1.4 Stochastic Bandits

Definition. A **stochastic bandit** is a collection of distributions $\nu = \{P_a : a \in \mathcal{A}\}$.

Here we are challenging to maximise the total reward $S_n = \sum_{t=1}^n X_t$ over n rounds.